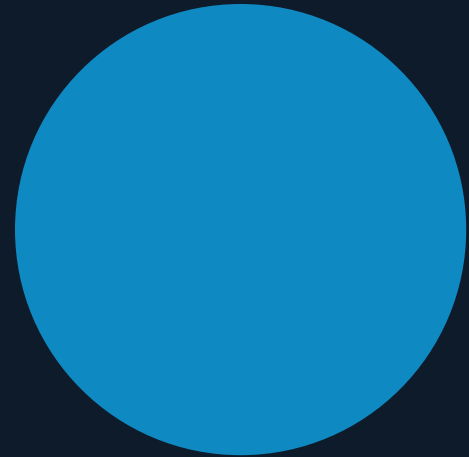


Enterprise Knowledge Intelligence Platform

Hybrid relational, document, and vector data architecture for
organizational knowledge management

B.Tech CSE (AI & Data Science) · KL University, Hyderabad

Updated with additional related papers



What this review covers

Literature review → gaps → proposed architecture → methodology

1

Problem & motivation

Why enterprise knowledge gets scattered and hard to retrieve

2

Literature review

Storage, semantic retrieval, security, and enterprise RAG papers

3

Critical analysis

What each research area solves and where it is incomplete

4

Gap analysis

Four gaps that define the EKIP problem space

5

Proposed system

Hybrid SQL + document + vector architecture

6

Methodology & evaluation

Search flow, permission filtering, and metrics

Why knowledge gets lost in organizations

Enterprise knowledge is spread across documents, wikis, chats, tickets, and file stores.

In many organizations, important knowledge exists, but users cannot easily find the right document at the right time.

1**Scattered content**

Knowledge is split across file shares, wikis, tools, and informal communication.

2**Keyword search weakness**

Exact-word search misses documents that use different words but have the same meaning.

3**Governance risk**

A search system must not reveal documents that the current user is not allowed to access.

EKIP goal: secure, flexible, meaning-aware knowledge retrieval.

Neither single-database nor simple wiki search is enough

The project needs storage flexibility, retrieval quality, and access governance together.

Relational-only system

Strong structure and permissions, but rigid schema and weak meaning-based search.



Document / wiki system

Flexible content storage, but usually keyword-heavy and coarse access control.



EKIP hybrid system

Combines governed access, flexible documents, and semantic vector retrieval.



Project statement: build a governed knowledge platform that stores enterprise content flexibly and retrieves it by meaning.

Literature Review 1: Data storage paradigms

What existing work says about SQL, document databases, and polyglot persistence.

Research area	Source	Main contribution	Limitation
Relational modeling	Kimball & Ross (2013)	Strong consistency, relationships, and structured querying.	Rigid for uneven documents.
Polyglot persistence	Sadalage & Fowler (2012)	Use different databases for different workloads.	Adds integration and consistency complexity.
Document databases	MongoDB documentation (2023)	Flexible JSON-like documents match evolving content.	Weaker joins than SQL.

Learning for EKIP: no single database is best for every knowledge-management workload.

Literature Review 2: Semantic search and vector retrieval

Search by meaning instead of only exact words.

1

Sentence-BERT — Reimers & Gurevych (2019)

Converts sentences into embeddings, so similar meaning becomes close vector distance.

2

HNSW — Malkov & Yashunin (2020)

Approximate nearest-neighbor graph makes large-scale vector search fast.

3

RAG — Lewis et al. (2020)

Retrieves relevant context first, then uses it for downstream language tasks.

Keyword search

same words

Semantic search

same meaning

Example: “employee vacation rules” can retrieve “leave policy” even when the exact words differ.

Learning for EKIP: semantic retrieval needs embeddings plus an efficient vector index.

Literature Review 3: Access control and API security

Enterprise search must return only permitted content.

RBAC

Ferraiolo, Kuhn & Chandramouli (2007)

Permissions are assigned by user role, such as admin, manager, or employee.

OAuth 2.0

IETF RFC 6749 — Hardt (2012)

Standard framework for delegated authorization in web APIs.

JWT

IETF RFC 7519 — Jones et al. (2015)

Compact token used to verify user identity and role claims.

Simple security flow: Login → JWT generated → API checks role → access allowed or denied

Additional related papers added

These strengthen the link between EKIP and enterprise-grade RAG/search systems.

Paper	Method / focus	Main finding	How it helps EKIP
FACTS About Building RAG-based Chatbots Akkiraju et al., 2024	Covers enterprise RAG pipeline design, access control, security, testing, latency, and references.	Enterprise RAG needs retrieval, security, testing, and performance together.	Directly supports secure enterprise semantic retrieval.
Optimizing & Evaluating Enterprise RAG Packowski et al., 2024	Shows practical enterprise RAG evaluation and importance of content design.	Enterprise RAG quality needs monitoring and evaluation, not only model choice.	Supports EKIP evaluation metrics and hybrid retrieval testing.
Secure Search in Enterprise Webs Bailey, Hawking & Matson, 2006	Studies document-level security during enterprise search.	Search must enforce document-level access before showing results.	Supports the G2 gap: retrieval must respect permissions.

Use these as 3 extra papers; do not add too many, because quality matters more than count.

Synthesis of the research strands

Each research area is strong alone, but incomplete for a full enterprise KMS.

Storage literature

- ✓ Structure and flexibility
- ✗ Does not solve retrieval quality or security

Retrieval literature

- ✓ Semantic matching and vector search
- ✗ Often assumes an open corpus

Security literature

- ✓ Authorization and stateless API security
- ✗ Does not organize or retrieve content

Critical conclusion

The gap is not absence of technology; the gap is secure integration of storage + search + access control.

Research gaps identified

Four gaps shape the EKIP design.

G1**No unified governance across content types**

Permissions are not consistently enforced across relational, document, and vector stores.

G3**Weak canonical linkage between stores**

A common document UUID is needed to link SQL metadata, document content, and vectors.

G2**Semantic retrieval often assumes an ungoverned corpus**

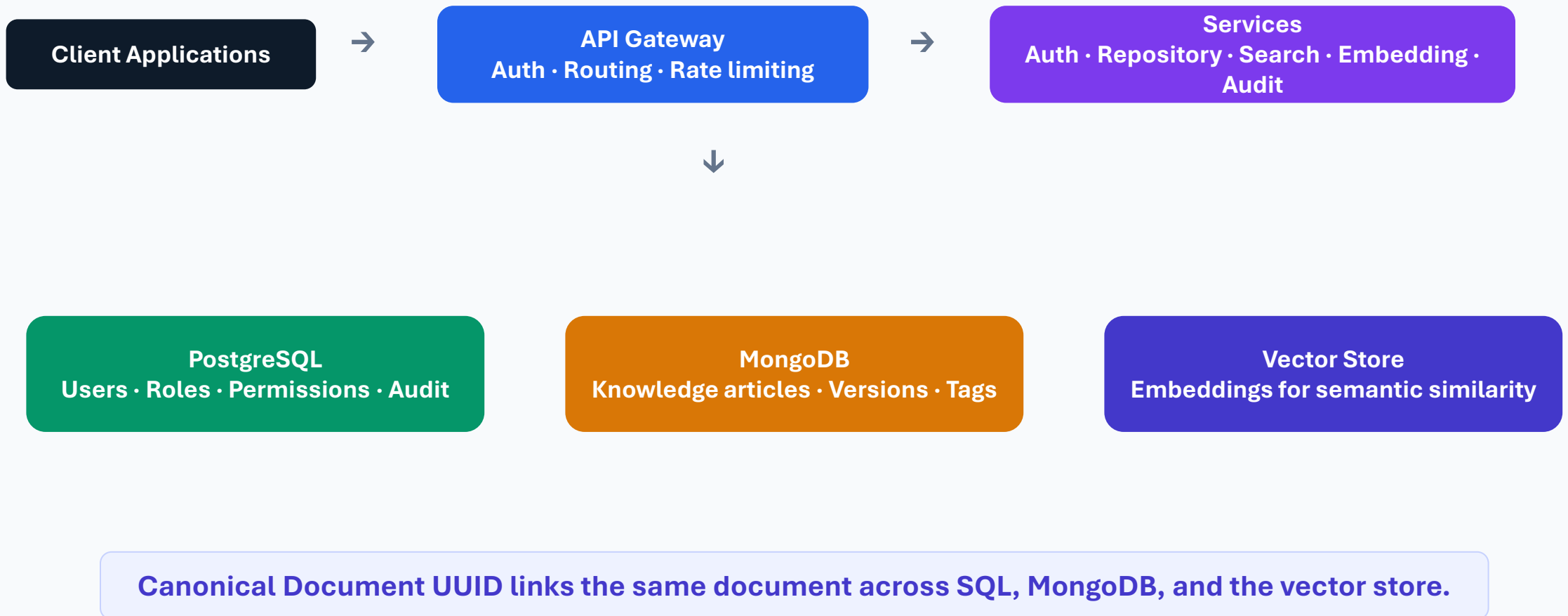
Vector search may retrieve relevant content before checking whether the user can view it.

G4**Limited hybrid retrieval evaluation**

Few systems compare keyword, semantic, and hybrid search in a governed setting.

EKIP hybrid architecture

A single secure API connects three specialized data stores.



Five functional modules

The system is divided by responsibility.

1

User & Role Management

RBAC, JWT authentication, salted password hashing

2

Knowledge Repository Management

Article CRUD, versioning, tagging, taxonomy

3

Document & Semantic Search

Chunking, embeddings, hybrid re-ranked retrieval

4

REST API & Authentication

OpenAPI-documented, OAuth 2.0 / JWT secured

5

Deployment, Testing & Docs

Containers, CI/CD, ADRs, monitoring

Technology stack

Each tool is selected for one clear system responsibility.

Backend

Node.js / NestJS or FastAPI

Relational DB

PostgreSQL

Document DB

MongoDB

Vector DB

Pinecone / Weaviate / pgvector

Embeddings

Sentence-Transformers

Auth

JWT + OAuth 2.0

Containers

Docker + Kubernetes

CI/CD

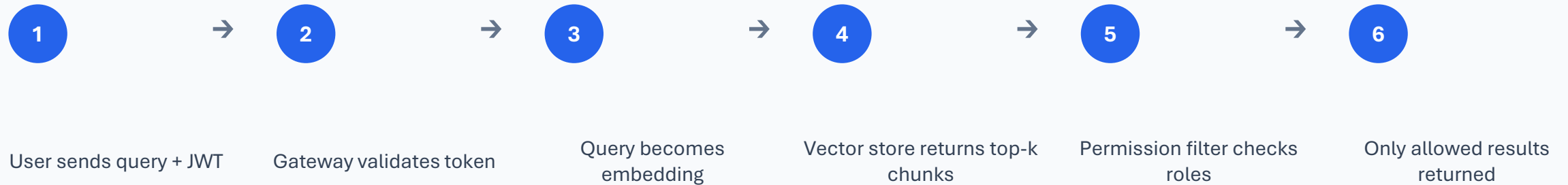
GitHub Actions

Monitoring

Prometheus + Grafana

Semantic search request flow

How one user query moves through EKIP.



Core methodology: retrieve by meaning first, then enforce access control before returning content.

DSA / DBMS angle in the project

Data structures and algorithms used inside the architecture.

1

Tables / indexes

PostgreSQL tables and indexes store users, roles, permissions, and audit records.

2

Document trees

MongoDB stores nested JSON-style documents for flexible knowledge articles.

3

Vector embeddings

Text chunks become high-dimensional numeric vectors.

4

ANN graph search

HNSW uses graph-based nearest-neighbor search for fast semantic retrieval.

This explains why the project is relevant to both DBMS and DSA: storage design + indexing + graph-based vector search.

Expected outcomes and metrics

How we will check whether EKIP works well.

1

Precision@k / Recall@k

Compare keyword-only, semantic-only, and hybrid search results.

2

API latency

Measure response time under concurrent requests.

3

Search throughput

Queries served per second as replicas scale.

4

Access compliance

100% of returned results must pass permission filtering.

Conclusion and future scope

EKIP closes the gap between governed access and meaning-aware retrieval.

EKIP unifies relational, document, and vector storage behind one secure REST API.

1

RAG Q&A

Automatic answers over the governed knowledge base

2

Connectors

Email, chat, ticketing, and enterprise repositories

3

Feedback re-ranking

Improve relevance using user interactions

4

Production evaluation

Test on larger real-world enterprise datasets

Selected references

Core papers, standards, and supporting sources.

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Thank You

Questions & Discussion

